THE GREAT DEBATE · THE FUTURE OF SRM

Quantum Computing

The most influential trend in security and risk management over the next five years.

Module: Security and Risk Management – Unit 12 Seminar

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OUR POSITION

The claim, and the obvious objection

Position

Quantum computing is the most influential SRM trend of 2026 to 2031.

Objection

A computer able to break today's encryption is likely five to ten years away.

Response

The machine and the impact are not the same thing. The impact is already here.

Why now: 1) attackers can harvest encrypted data today and decrypt it later; 2) migration is a decade-long programme that has already begun.

SCOPE OF THE THREAT

A threat to every sector

Government

Classified and citizen records

Healthcare

Long-lived patient data

Banking & finance

Payments and settlement

Corporate

Intellectual property

Harvest now, decrypt later. Encrypted data copied today can be stored cheaply and unlocked once a quantum computer exists. Anything that must stay secret for years is exposed now, breaking confidentiality, integrity and authentication.

WHY THE TIMING MATTERS

The next five years are decisive

2024
NIST publishes quantum-safe standards

2028
Discovery: know where you use encryption

2031
Critical systems migrated

2035
Full migration complete

The window this debate is about

Sources: NIST (2024); NCSC (2025); BIS (Auer et al., 2025). The work is not in the future; it runs from now to 2031.

METHODOLOGY

The way we assess this risk has to change

The old approach falls short

- Likelihood multiplied by impact, and ALE built from a rate of occurrence.
- The arrival date of a quantum computer is unknown, so the rate is a guess.
- A precise figure with no data behind it is false confidence (Cox, 2008).

The updated approach

- Risk as activity, consequences and uncertainty (Aven and Thekdi, 2025).
- Report the strength of knowledge, not just a single number.
- Use scenarios, and state plainly what is known and what is not.

MITIGATION

How we act, starting now

Inventory your cryptography

Build a full picture of where encryption is used. You cannot protect what you cannot see.

Build crypto-agility

Design systems so an algorithm can be swapped without rebuilding everything.

Run hybrid encryption

Combine classical and post-quantum algorithms, so security holds if either survives.

Adopt the new standards

Migrate to ML-KEM, ML-DSA and SLH-DSA (FIPS 203, 204, 205) as products support them.

And the supply chain: a bank must track whether its core banking partners are becoming quantum-ready. You cannot do this alone.

WIDER RESPONSIBILITIES

Legal, social, ethical and professional issues

Legal

UK GDPR requires state-of-the-art security. Old encryption for sensitive data becomes hard to defend, and a breach may begin at the moment data is stolen.

Social

A shared problem. Large banks and hospitals move slowly, and smaller firms depend on their vendors to deliver the fix.

Ethical

Be honest about uncertainty. Do not overstate the threat to win budget, and do not use that uncertainty as an excuse to delay.

Professional

The supply chain cannot be fixed alone. When mitigation is too slow, harm follows, as hospitals hit by ransomware have shown.

OUR CASE

Already on the risk register

- The machine is not here yet, but the work is.
- The risk is live now, because of harvest now, decrypt later.
- The deadlines are already set: 2028, 2031, 2035.
- And the way we assess the risk has to change.

Quantum computing is the most influential SRM trend of the next five years.

REFERENCES

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